CertiVar: Prevalence-Free Certified Assay Likelihood Ratios for Missense Variant-Effect Predictor Calls

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Abstract

Variant-effect predictors are a common source of evidence for missense variants of uncertain significance (VUS), and the Bayesian form of the ACMG/AMP guidelines expresses evidence strength as a likelihood ratio. Gene-specific calibrations estimate this ratio from few labelled variants, as point estimates or resampling-based bounds that are not designed to remain valid once a threshold has been chosen from the same data. We present CertiVar, which turns a predictor score into loss-of-function (LoF) calls, functional calls or abstentions using class-conditional conformal thresholds learned from saturation genome editing or deep mutational scanning data, with one variant per residue. It then certifies a bound on the likelihood ratio of each call type with respect to the assay outcome that stays valid after a threshold is chosen from a grid, by a Bonferroni correction, or certifies evidence levels by fixed-sequence testing. Because these guarantees condition on the assay class, none needs the unknown share of LoF variants among the variants being called, provided their within-class score distribution matches that of the assayed variants. On 14,477 assayed missense variants in BRCA1, BRCA2, BAP1, RAD51C and MSH2, scored by AlphaMissense, REVEL and ESM-2, mean worst-class coverage over 50 splits was 0.896–0.908 at a 0.90 target in all 15 gene–predictor pairs and stayed at 0.89 or above when the LoF share of the calibration data ranged from 5% to 50%; marginal conformal prediction calibrated at a 5% share covered LoF variants at only 0.11 in its worst pair. In resampling experiments with known population likelihood ratios, all certified bounds of a draw held in 99.8% of draws (guaranteed: at least 90%), against 56.5% for percentile-bootstrap bounds, and fixed-sequence testing, which certifies one evidence level at a time, more than doubled the certification of moderate LoF evidence. The certificates concern the assay, not pathogenicity; certified evidence was mostly supporting-level and held for 88.8% of certified bounds in held-out protein regions, below the nominal 95% per call type. Code: `https://github.com/aaryavvatts-lab/certivar`

Keywords: variant effect prediction; conformal prediction; likelihood ratio; ACMG/AMP evidence calibration; saturation genome editing; deep mutational scanning; variants of uncertain significance; multiple testing.

1 Introduction

In the five cancer-predisposition genes studied here, 81% of the ClinVar missense records classified as (likely) pathogenic, (likely) benign or uncertain are variants of uncertain significance (VUS; our count from the 26 October 2026 release [1]), and for many of them computational predictors are among the few sources of evidence [2]. The guidelines of the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) grade evidence qualitatively [3]; their Bayesian adaptation equates each strength with a likelihood ratio, the odds of pathogenicity (OddsPath) [4, 5]. Calibration workflows report this ratio as a point estimate or a resampling-based bound [2, 6, 7]. For a single gene, few labelled variants are available and the threshold that defines the evidence is chosen from the same data; these bounds are not designed to remain valid after this choice.

Two features of the data make the estimate fragile. First, the share of damaging variants differs between calibration sets and the variants being classified, and is unknown for the latter: in our genes, 45% of confidently labelled ClinVar missense variants are pathogenic, whereas 6.0% to 25.9% of the other assayed missense variants lose function in saturation genome editing (SGE) or deep mutational scanning (DMS) experiments (Table 1). Second, gene-level labelled sets are small and substitutions at one residue are dependent: in a resampling design with known truth, percentile-bootstrap bounds were respected in only 76.5% of threshold selections (Section 4.4).

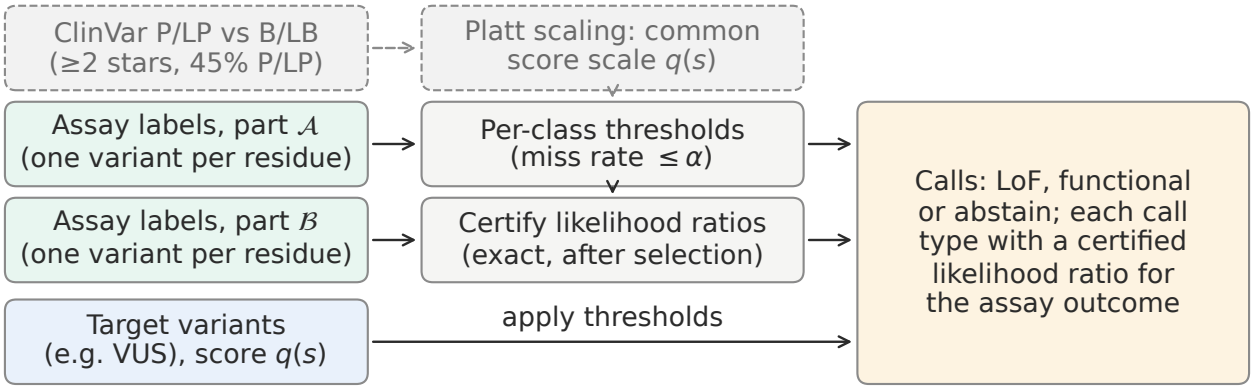


Figure 1: CertiVar. Per-class thresholds are set on part $\mathcal{A}$ of a gene's assayed residues, the likelihood ratio of each call type with respect to the assay outcome is certified on a disjoint part $\mathcal{B}$, and the thresholds are applied to the target variants. The grey path (Platt scaling on ClinVar labels) only puts predictors on a common scale and does not change any call.

We introduce CertiVar (Figure 1), which turns a predictor score into loss-of-function (LoF), functional or no calls. Per-class (Mondrian) conformal thresholds set from SGE or DMS labels bound how often each class is mis-called, and a certification step bounds the likelihood ratio of each call type with respect to the assay outcome, valid after a threshold has been selected; neither needs an estimate of the LoF share. The certified quantity is how strongly a call predicts the assay result; it is evidence about pathogenicity only to the extent that the assay is clinically validated. Our contributions: (1) a formulation of predictor-based calling as abstaining classification on residue-clustered data, with proofs that the certified likelihood-ratio bounds hold after threshold selection, that per-class coverage after selection is at least one minus the largest grid level, and that a fixed-sequence variant needs no multiplicity correction over the grid for one evidence level (Section 3); (2) a known-truth benchmark in which exact binomial bounds held where bootstrap bounds and point estimates did not (Section 4.4); (3) evidence on five genes and three predictors that per-class coverage stays at or near the target when the calibration LoF share moves from 5% to 50%, where marginal and prior-reweighted conformal prediction fail (Section 4.2); and (4) where the guarantees stop (Section 4.5).

2 Related work

ClinGen maps predictor scores to ACMG/AMP evidence strengths through local likelihood ratios estimated against ClinVar labels, with a one-sided bootstrap bound [2]; the approach has been extended to more predictors [6], to assay scores [5, 8, 9] and to gene- and domain-specific calibration [7, 10, 11], with uncertainty from the bootstrap [2, 6] or sign-test bounds on resampled likelihood ratios [7]. Split conformal prediction gives finite-sample coverage [12]; its class-conditional (Mondrian) form controls coverage within each class [12, 13]; order-statistic thresholds give training-conditional control [14, 15]; keeping one member per group handles clustered data [16]; and abstaining classifiers trade the share of inputs classified against error [17]. Selecting a threshold from a grid with binomial bounds and a multiplicity correction, or by fixed-sequence testing, is the template of selective classification with guaranteed risk, risk-controlling prediction sets and Learn-then-Test [18–20]. Harrizi et al. reported split-conformal prediction sets for missense predictors on ClinVar benchmarks at 90% coverage [21]; CertiVar instead controls coverage within each class, calibrates on the gene's assay labels and certifies the likelihood ratio of the calls. Large-sample intervals for a ratio of binomial proportions [22, 23] are narrower than our ratio of exact endpoints but carry no finite-sample guarantee. Label-shift methods adjust posteriors or conformal weights to an estimated target prior [24–26]; class-conditional quantities need no prior, provided the target satisfies class-conditional invariance (Assumption 2). Relative to these methods, CertiVar contributes a certificate on the OddsPath scale for both call types of an abstaining rule on residue-clustered assay labels, with coverage after threshold selection, a fixed-sequence variant, and a known-truth benchmark of how often point estimates and bootstrap bounds overstate evidence.

3 Method

Setting. Each missense variant has a predictor score s, oriented so that higher means more damaging, and an assay label $Y \in \{0, 1\}$, with $Y = 1$ for loss of function (LoF). A *residue* is a protein position of one gene; the substitutions at a residue share structural context and form one cluster. A gene's assayed missense variants are split by residue into a calibration part $\mathcal{A}$ and a certification part $\mathcal{B}$; the variants to be called (for example VUS) form the target. We write $q(s) \in [0, 1]$ for the score after Platt scaling [27] fitted to confidently labelled ClinVar variants (pathogenic or likely pathogenic, P/LP, against benign or likely benign, B/LB); $q(s)$ estimates the probability that such a ClinVar variant is pathogenic, not that an assay outcome is LoF. Because q is increasing in s, every procedure below makes the same calls on q as on s; q only puts predictors on one scale (the baselines use its values), so no ClinVar label affects a CertiVar call.

Per-class thresholds. The nonconformity of label 1 is $r_1 = 1 - q$ and of label 0 is $r_0 = q$. From each residue in $\mathcal{A}$ we keep one variant chosen uniformly at random. For class k, let n_k be the number of kept variants with label k and $r^{(k)}_{(1)} \le \cdots \le r^{(k)}_{(n_k)}$ their sorted nonconformities. The threshold is $\hat{t}_k = r^{(k)}_{(j_k)}$ (∞ if $j_k > n_k$), where j_k is the in-expectation rank or the probably approximately correct (PAC) rank ($j_k = n_k + 1$ if no j qualifies):

$$j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil \text{ (in expectation)} \quad \text{or} \quad j_k = \min\{\, j : \; F_{\mathrm{Beta}(j,\, n_k - j + 1)}(1 - \alpha_k) \le \delta \,\} \text{ (PAC).} \tag{1}$$

The prediction set is $C(q) = \{k : r_k \le \hat{t}_k\}$; CertiVar calls LoF if $C = \{1\}$, functional if $C = \{0\}$, and abstains otherwise. A functional call on a LoF variant requires $1 \notin C$, so per-class coverage bounds both error rates.

Certified likelihood ratios. The ACMG/AMP guidelines [3] grade evidence in qualitative categories; in their Bayesian adaptation, supporting, moderate, strong and very strong evidence towards pathogenicity correspond to a likelihood ratio above 2.1, 4.3, 18.7 and 350, and supporting, moderate and strong evidence towards benignity to one below 0.48, 0.23 and 0.053 [4, 5]; these values were derived for a prior probability of pathogenicity of 0.10, but the ratios themselves need no prior. We use these values as levels and write, for example, *LoF-supporting* for LoF calls whose likelihood ratio exceeds 2.1. Our likelihood ratios are defined with respect to the assay outcome Y and averaged over all variants that receive a call, as OddsPath is for a functional assay [5]; they are not the local, score-specific ratios that ClinGen uses for the computational criteria PP3/BP4 [2]. A level thus describes the call class as a whole, need not hold for each call, and must not be combined with the same assay's functional evidence (PS3/BS3) or with PP3/BP4 from the same predictor. For thresholds t, the call rates (probabilities, not counts) are $\mathrm{TP}(t) = \Pr(\text{call LoF} \mid Y = 1)$, $\mathrm{FP}(t) = \Pr(\text{call LoF} \mid Y = 0)$, $\mathrm{FN}(t) = \Pr(\text{call functional} \mid Y = 1)$ and $\mathrm{TN}(t) = \Pr(\text{call functional} \mid Y = 0)$, and $\mathrm{LR}^+ = \mathrm{TP}/\mathrm{FP}$, $\mathrm{LR}^- = \mathrm{FN}/\mathrm{TN}$.

For each α on a grid G we compute thresholds on $\mathcal{A}$ ($\alpha_0 = \alpha_1 = \alpha$), call one variant per residue of $\mathcal{B}$, and bound the four rates by one-sided Clopper–Pearson bounds [28] at level $\epsilon = \delta/(2|G|)$, written ℓ (lower) and u (upper): for x calls among n variants of a class, $\ell = F^{-1}_{\mathrm{Beta}(x,\, n-x+1)}(\epsilon)$ (0 if $x = 0$) and $u = F^{-1}_{\mathrm{Beta}(x+1,\, n-x)}(1 - \epsilon)$ (1 if $x = n$, including $n = 0$). The certified bounds are $L^+(\alpha) = \ell_{\mathrm{TP}}/u_{\mathrm{FP}}$ and $U^-(\alpha) = u_{\mathrm{FN}}/\ell_{\mathrm{TN}}$ ($U^- = \infty$, never meeting a level, if $\ell_{\mathrm{TN}} = 0$; $u_{\mathrm{FP}} > 0$ always; for point estimates, $x/0 = \infty$ for $x > 0$, and $0/0$ is undefined and never selected). For each level, Algorithm 1 returns the α that calls the most variants among those whose certified bound meets the level. Because $\hat{t}_0$ and $\hat{t}_1$ both decrease weakly in α, no variant can be called LoF at one α and functional at another (Appendix S1). Line 7 also requires, for functional calls, α at least as large as the LoF choice of the same strength; this is a reporting convention, not needed for validity, and can only lower functional yields. The *fixed-sequence* variant replaces the Bonferroni level $\delta/(2|G|)$ by $\delta/2$ and tests the grid in an order fixed in advance, stopping at the first α that does not meet the level (Theorem 2). We test LoF levels in increasing and functional levels in decreasing α, because few variants are called functional at small α. All thresholds and bounds cost $O(|G|\, n \log n)$ for n residues, about 5 ms per gene on a laptop CPU.

Algorithm 1 CertiVar for one gene and one evidence level

Require: scores q; assay labels split by residue into $\mathcal{A}, \mathcal{B}$; grid G; δ; targets τ (LoF) and τ' (functional)

1: keep one variant per residue of $\mathcal{A}$ and of $\mathcal{B}$
2: **for** $\alpha \in G$ **do**
3: thresholds $\hat{t}_0, \hat{t}_1$ on the kept variants of $\mathcal{A}$ by (1); call the kept variants of $\mathcal{B}$
4: $L^+(\alpha) \leftarrow \ell_{\mathrm{TP}}/u_{\mathrm{FP}}$, $U^-(\alpha) \leftarrow u_{\mathrm{FN}}/\ell_{\mathrm{TN}}$ (Clopper–Pearson, level $\delta/(2|G|)$ each)
5: **end for**
6: $\hat{\alpha}^+ \leftarrow \arg\max_{\alpha \in G}\{\text{share called LoF on } \mathcal{B} : L^+(\alpha) > \tau\}$
7: $\hat{\alpha}^- \leftarrow \arg\max_{\alpha \in G}\{\text{share called functional on } \mathcal{B} : U^-(\alpha) < \tau',\ \alpha \geq \hat{\alpha}^+\}$ (over all $\alpha \in G$ if $\hat{\alpha}^+$ does not exist)
8: **output** LoF calls at $\hat{\alpha}^+$ and functional calls at $\hat{\alpha}^-$ with their certified levels (no calls of a type whose set is empty); abstain otherwise

3.1 Guarantees

Let Q be the law of a variant obtained by drawing a residue from the gene's population of assayed residues and then one of its variants uniformly at random (binary-labelled, non-splice variants outside the ClinVar labelled set).

Assumption 1 (Clustered sampling)**.** Residues in $\mathcal{A}$, in $\mathcal{B}$ and of a test variant are independent draws from one residue population; within a residue, the kept or tested variant is chosen uniformly at random; q does not depend on these data.

Assumption 2 (Class-conditional invariance)**.** Target variants have, for $k \in \{0, 1\}$, the same law of q given $Y = k$ as a test variant under Assumption 1; their LoF share $\pi = \Pr(Y = 1)$ may differ.

Proposition 1 (Per-class coverage)**.** *Under Assumption 1, for a fixed α_k, with the in-expectation rank and conditionally on n_k, $1 - \alpha_k \leq \Pr(k \in C \mid Y = k, n_k) \leq 1 - \alpha_k + 1/(n_k + 1)$, the upper bound holding when there are no ties.*

Proposition 2 (High-probability per-class coverage)**.** *Under Assumption 1, for a fixed α_k and with the PAC rank, $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \alpha_k$ with probability at least $1 - \delta$ over the calibration data for each k, and for both classes with probability at least $1 - 2\delta$.*

Proposition 3 (Coverage after selection)**.** *Under Assumption 1, for any $\hat{\alpha} \in G$ chosen from data independent of the test variant (for example by Algorithm 1), the in-expectation rank gives $\Pr(k \in C \mid Y = k) \geq 1 - \max G$; the PAC rank at level $\delta/|G|$ gives $\Pr(k \in C \mid Y = k, \hat{t}_k) \geq 1 - \hat{\alpha}$ for each k with probability at least $1 - \delta$.*

Theorem 1 (Certified likelihood ratios)**.** *Under Assumption 1, conditionally on $\mathcal{A}$ and with probability at least $1 - \delta$ over $\mathcal{B}$, $\mathrm{LR}^+(\hat{t}(\alpha)) \geq L^+(\alpha)$ under Q for every $\alpha \in G$ simultaneously; the same holds for $\mathrm{LR}^- \leq U^-$ with a second δ. Because the event does not depend on the level, each call type returned by Algorithm 1 reaches its certified level at every evidence level reported, with probability at least $1 - \delta$ (both call types with probability at least $1 - 2\delta$), whichever α is selected.*

Theorem 2 (Fixed-sequence certification)**.** *Under Assumption 1, fix a level τ and an order of G chosen in advance. Reject $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \leq \tau$ when $\ell_{\mathrm{TP}}/u_{\mathrm{FP}} > \tau$ with both endpoints at level $\delta/2$, stop at the first α not rejected, and select the rejected α (a prefix of the order) with the largest yield. Then, conditionally on $\mathcal{A}$, the selected LoF calls have $\mathrm{LR}^+ > \tau$ with probability at least $1 - \delta$. The same holds for functional calls with $u_{\mathrm{FN}}/\ell_{\mathrm{TN}} < \tau'$.*

Corollary 1 (No prior needed)**.** *Under Assumptions 1 and 2, Propositions 1–3 and Theorems 1–2 hold for target variants whatever their LoF share $\pi \in (0, 1)$: for a fixed α, LoF variants are called functional at rate at most α_1, and the certified likelihood ratios with respect to the assay outcome need no prior.*

Propositions 1 and 2 are the class-conditional and training-conditional conformal guarantees [12–15], here for residue-clustered data. Proposition 3 holds because both ranks in (1) are non-increasing in α, so coverage at any selected α is at least coverage at $\max G$. Theorem 1 follows the risk-control template [19]: given $\mathcal{A}$ and the labels in $\mathcal{B}$, the four counts are binomial, and a union bound over the $2|G|$ endpoints covers every α, including a selected one. Theorem 2 follows Learn-then-Test [20]: a true H_α is rejected only if an endpoint fails, and a false claim requires rejecting the first true hypothesis in the order. Proofs are in Appendix S1.

Table 1: Datasets. *ClinVar labelled*: missense single-nucleotide variants (SNVs) with ≥2 review stars and a P/LP or B/LB call (used to fit the common score scale and, in Table S4 only, for an exploratory check) and the number that are P/LP. All shares are variant-level (descriptive). *Assayed*: missense SNVs with a binary assay label that are not in the ClinVar labelled set, their residues (Res.) and their LoF share; *assayed VUS*: the subset classified VUS in ClinVar. Assay labels are the authors' published classes (authors) or, where none are deposited, derived from the assay's own synonymous and nonsense controls (controls).

			ClinVar labelled		Assayed			Assayed VUS	
Gene	Assay	Labels	n	P/LP	n	Res.	LoF	n	LoF
BRCA1	SGE [29]	authors	345	156	1,617	307	0.161	272	0.173
BRCA2	SGE [30]	controls	309	94	3,435	676	0.130	704	0.207
BAP1	SGE [31]	authors	14	6	4,294	696	0.213	1,069	0.166
RAD51C	SGE [32]	authors	13	10	2,225	355	0.259	550	0.396
MSH2	DMS [33]	authors	178	121	2,906	893	0.060	1,542	0.067
Total			859	387	14,477	2,927		4,137	

Scope. After selection, per-class coverage is guaranteed only at $1-\max G$ with the in-expectation rank; we report the coverage achieved in Section 4.4. Theorem 2 certifies one level at a time and gives no numerical bound. Assumption 1 treats assayed residues as exchangeable, which a contiguous region need not be; Assumption 2 holds only approximately for VUS, on which AlphaMissense discriminates less well (Section 4.1), and not for variants whose assay result would be intermediate.

4 Experiments

4.1 Data and protocol

We took every GRCh38 missense single-nucleotide variant in BRCA1, BRCA2, BAP1, RAD51C and MSH2 that is in ClinVar [1] (release of 26 October 2026) or was assayed, annotated each on its MANE Select transcript with the Ensembl Variant Effect Predictor [34] (AlphaMissense [35], REVEL [36]) and computed ESM-2 650M [37] log-likelihood ratios by wild-type marginals [38]. Labels come from five MaveDB [39] score sets (Table 1; accessions in Appendix S5): the authors' classes for BRCA1, BAP1 and RAD51C, the authors' threshold for MSH2 (positive loss-of-function score [33]), and, for BRCA2, which deposits no classes, Gaussians fitted to the assay's synonymous and nonsense variants (LoF if the posterior is at least 0.95, functional if at most 0.05). Because this control rule disagreed with the authors' classes for 216 of the 4,023 BAP1 variants confident under both, we repeat the main experiments under alternative label rules (Section 4.5). We excluded the 729 assayed variants in splice regions and set aside the 1,274 intermediate variants. One Platt map per predictor, pooled over genes, used the 859 ClinVar missense variants with at least two review stars and a P/LP or B/LB call (45.1% P/LP), which are excluded from every assay-based evaluation set, leaving 14,477 assayed variants with a binary label. Pooled over genes, AlphaMissense has an area under the ROC curve (AUROC) of 0.946 on the ClinVar set, 0.840 on the other assayed variants and 0.779 on assayed VUS (non-overlapping 95% residue-bootstrap intervals). Baselines are pooled (marginal) split conformal prediction; label-shift-weighted conformal prediction [26] with the true target share or its expectation–maximization (EM) estimate [24]; fixed tiers ($q \geq 0.9$ LoF, $q \leq 0.1$ functional); the published AlphaMissense classes (> 0.564, < 0.34); and the REVEL PP3/BP4 thresholds (≥ 0.644, ≤ 0.290) [2]. None claims per-class coverage, and the published cut-offs target pathogenicity, so we judge them only as assay classifiers. Coverage is residue-weighted as in Q; *worst-class coverage* is the smaller of the mean LoF and mean functional coverages over repetitions. Every split is by residue (designs in Appendix S6).

4.2 Per-class coverage, and a shift in the calibration LoF share

Over 50 random halvings of each gene's residues (calibrate on one half, test on the other), the in-expectation rule at $\alpha = 0.1$ gave worst-class coverage between 0.896 and 0.908 in all 15 gene–predictor pairs, none more than one Monte Carlo standard error (0.002–0.010) below the target (Figure 2A; per-pair values in

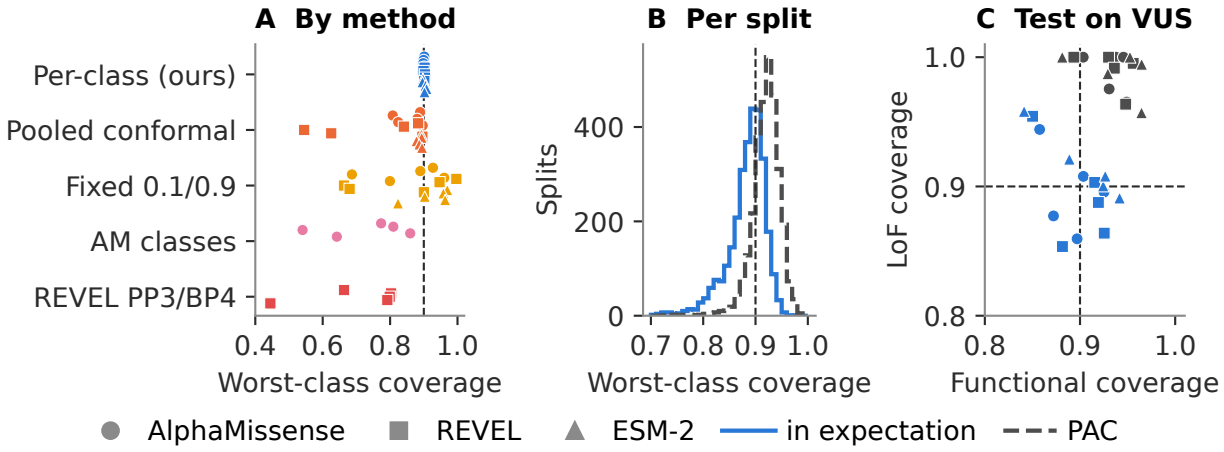


Figure 2: Per-class coverage at $\alpha = 0.1$; markers give the predictor in panels A and C. (A) Worst-class coverage for each gene–predictor pair (50 splits); AM classes: the published AlphaMissense classes. (B) Smaller of the two per-class coverages in each split, pooled over pairs (3,000 splits per rule). (C) Coverage on assayed VUS with thresholds set on non-VUS variants at other residues (50 repetitions); in C, blue: in expectation, grey: PAC.

Table S1). Pooled conformal fell to 0.545 on LoF variants and below 0.85 in 5 pairs, because a marginal threshold controls coverage averaged over the two classes, not within each; the published AlphaMissense and REVEL cut-offs covered as little as 0.540 (BAP1) and 0.445 (MSH2) of functional variants. The per-class rule abstained on 35% of variants on average (17–47%), against 42% for pooled conformal and 11% for the AlphaMissense classes, which trade coverage for decisiveness. Calibrating on all variants instead of one per residue used five times as many labels and abstained less (30.4% against 35.2%), but worst-class coverage fell below 0.90 in 13 of 15 pairs (4 of 15, lowest 0.897, for one variant per residue in that run).

To test whether per-class thresholds need a prevalence estimate, we subsampled each calibration half to a LoF share of 0.05 to 0.5 and tested at the natural share (100 repetitions; Figure 4A). The per-class rule kept worst-class coverage between 0.897 and 0.925 in every pair at every share (0.891–0.925 in a separate run balanced at 50% LoF), with abstention rising from 37% to 83% as LoF variants became scarce. Pooled conformal calibrated at share 0.05 covered LoF variants at 0.113 in its worst pair, and label-shift-weighted conformal fell to 0.53–0.60 with the true target share and to 0.04 with the EM estimate, which missed the LoF share by 0.255 on average because ClinVar pathogenicity and assay loss of function differ in definition as well as in class share.

4.3 Reliability across splits and a real change of population

With 17 to 93 LoF residues per calibration half, a single split can miss the in-expectation guarantee: over 200 splits per pair, LoF coverage was below 0.90 in 36.4% of splits and functional coverage in 43.6% (Figure 2B). PAC thresholds ($\delta_P = 0.1$) reduced these to 7.9% and 10.8% (test-half estimates of the coverage that Proposition 2 bounds) at the price of abstaining on 58.4% of variants against 35.4%. Calibrated on non-VUS variants of one half of each gene's residues and applied to the assayed VUS of the other half (residue-weighted LoF share 17.0% in calibration, 19.8% in VUS), in-expectation thresholds kept LoF coverage between 0.854 and 0.958 and functional coverage between 0.841 and 0.942, each below 0.90 in 7 of 15 pairs (Figure 2C); PAC thresholds stayed at or above 0.957 (LoF) and 0.881 (functional) but abstained on 81.7% of VUS.

4.4 Certified likelihood ratios

Treating each gene's assayed residues as the population (the oracle design), we drew calibration (30%) and certification (50%) residues with replacement, 100 times per pair, ran Algorithm 1 with $\delta = 0.05$ and $G_4 = \{0.05, 0.10, 0.20, 0.30\}$, and computed the true residue-weighted likelihood ratio of every selection on the whole population. We compared Clopper–Pearson bounds without multiplicity correction, a one-sided

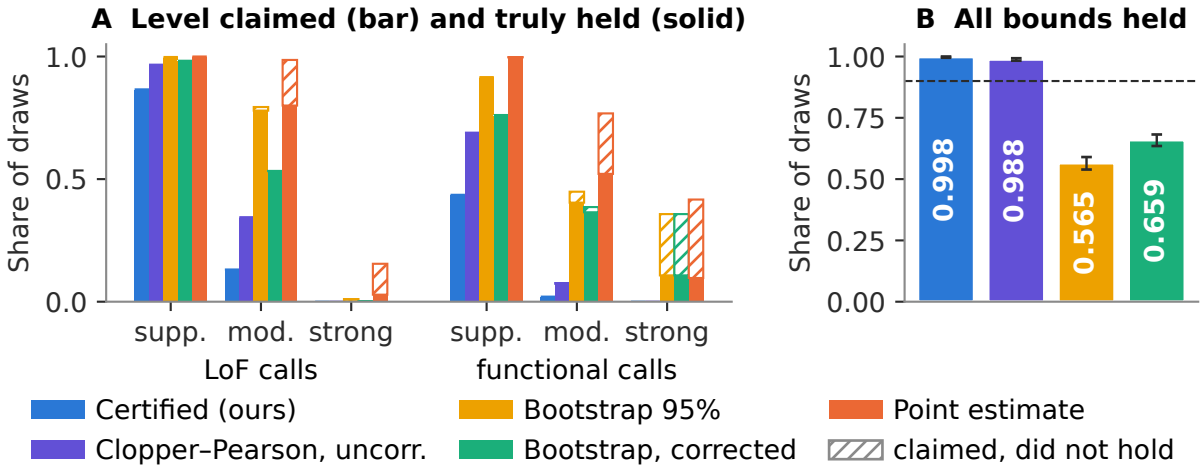


Figure 3: Threshold selection judged against the population likelihood ratio (oracle design, 1,500 draws). (A) Share of draws in which each rule claims a level (full bar) and in which it held (solid). (B) Share of draws in which every claimed bound was respected, with 95% intervals from resampling draws (a point estimate is not a bound and is omitted); dashed line at the guaranteed 0.90.

95% percentile bootstrap (2,000 resamples), the same bootstrap at level $\delta/(2|G|)$, and the point estimate, all applied to the same data. Every bound claimed in a draw was respected in 99.8% of the 1,500 draws for the certified rule (Theorem 1 guarantees at least 90% for both call types together and 95% for each), 98.8% for uncorrected bounds, 56.5% for bootstrap bounds and 65.9% for the corrected bootstrap (Figure 3B); per selection, the bootstrap bound was respected in only 76.5%. Most bootstrap failures came from certification samples with no false call of that type, where the bound failed 89% of the time. Claimed levels held in all 2,181 certified claims, but in only 91.0% of bootstrap and 79.7% of point-estimate claims (Figure 3A). On this grid a binomial confidence bound on one variant per residue (exact here; Appendix S7 adds Katz intervals), not the multiplicity correction, separates valid rules from the others; the correction costs yield (LoF-moderate certified in 13.1% of draws with it, 34.5% without).

CertiVar certified LoF-supporting calls in 86.6% of draws (calling 25.8% of variants), functional-supporting calls in 43.6% (61.1% called), LoF-moderate in 13.1% and functional-moderate in 2.1%, and never a strong level. In a fresh set of 1,500 draws, per-class coverage at the selected α averaged 0.82 (LoF) and 0.80 (functional), above the guaranteed 0.70 (Proposition 3): selecting for evidence strength trades coverage for confidence. Fixed-sequence testing, which certifies one level at a time, raised LoF-moderate certification in these draws from 12.9% (Bonferroni) to 29.1% of draws and functional-supporting from 42.5% to 60.6% (Figure 4B; with an all-levels guarantee it trades supporting for moderate evidence, Appendix S7), with 99.7% of 2,705 claims holding; on a 20-point grid G_{20} (0.02 to 0.40) it kept LoF-moderate at 26.7% where the Bonferroni rule fell to 6.2%. The testing order matters: functional-supporting, tested in increasing α, was certified in only 25.7% of draws (60.6% in decreasing order). On held-out fifths of the residues, empirical likelihood ratios respected the certified bounds in 98.7% of 2,091 cases (bootstrap: 76.8%); on VUS at other residues, LoF-supporting held in all 443 certified claims and LoF-moderate in 25 of 28 (89%, below the nominal 95%). Against ClinVar labels (exploratory, not independent of the assays), LoF calls had likelihood-ratio lower bounds of 4.9–19.6 (BRCA1), 7.6–9.4 (BRCA2) and 2.6–3.2 (MSH2) (Table S4).

4.5 How much data, from where, and with which labels

Calibration data had to come from the same gene: calibrating on the other four genes left 8 of 15 pairs below 0.85 (worst 0.487), consistent with gene-specific evidence strength [10, 11]. Calibrating on three contiguous quarters of a protein and testing on the fourth gave worst-class coverage of 0.860–0.923, and the true ratio respected 88.8% of 1,891 certified bounds, below the nominal 95%. With 150 calibration residues drawn at random, coverage averaged over genes met the target with 40.5% abstention (82.8% with 25 residues; Figure S2). Coverage was lower on the third of variants nearest the assay class boundary (0.865

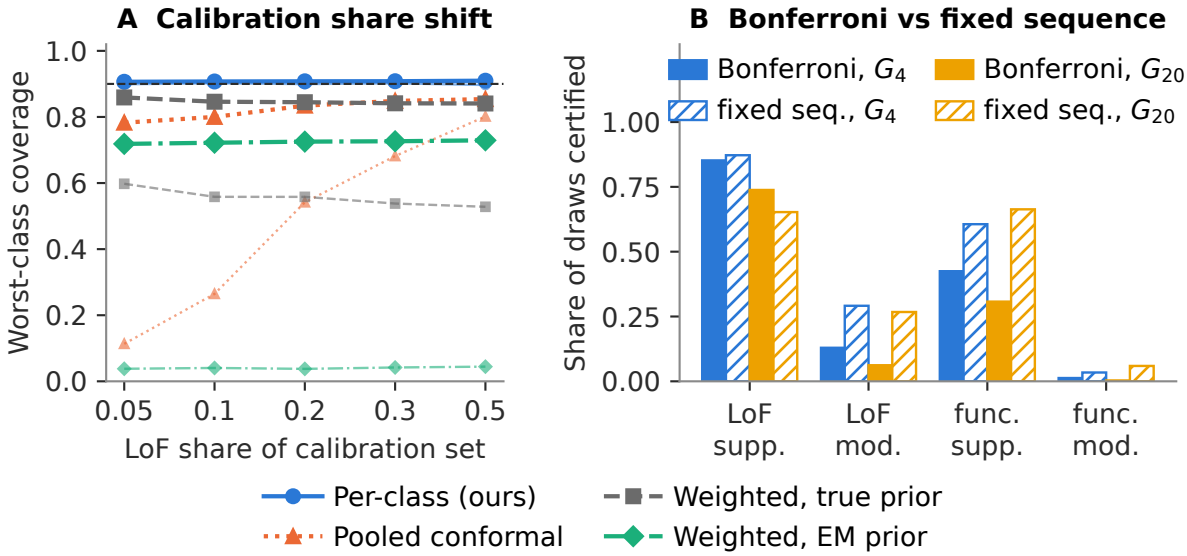


Figure 4: (A) Worst-class coverage when the calibration half is subsampled to a given LoF share (evenly spaced axis): mean over 15 pairs (thick) and worst pair (thin). (B) Share of oracle draws certifying each level with a Bonferroni correction or fixed-sequence testing, on the four-point (G_4) and 20-point (G_{20}) grids.

LoF, 0.873 functional). Intermediate variants, which no guarantee covers, were called LoF in 19.6% and functional in 36.6% of cases; on 1,534 unassayed BRCA1 VUS, AlphaMissense thresholds abstained on only 53, which suggests a shifted score distribution. Relabelling BRCA2, BAP1 and RAD51C with alternative rules (Table S3) changed yields but not validity: the lowest per-class coverage over predictors stayed between 0.894 and 0.902, and certified bounds held in 2,530 of 2,533 oracle selections.

5 Discussion

Thresholds selected on point estimates or percentile-bootstrap bounds often claimed more evidence than held. Because these are call-averaged ratios, not the local ratios of ClinGen calibrations, this does not show that published PP3/BP4 thresholds are wrong, only that point estimates and bootstrap bounds can overstate evidence at single-gene sample sizes. CertiVar makes the claim checkable without an estimate of the share of damaging variants. Its intended use is a gene with a functional assay: calling variants the assay missed within assayed regions, and checking whether a predictor's claimed evidence strength holds against the assay. Appendix S7 counts the ClinVar VUS that receive certified calls, tests the invariance assumption on unlabelled targets and adds Katz and pseudocount-bootstrap baselines. The cost is conservatism: certified evidence was mostly supporting-level, and PAC thresholds abstained on 58% to 82% of variants.

Limitations. The labels are assay classes, not clinical pathogenicity [5]; clinical use requires an assay validated against clinical classifications, as its authors did for BRCA1 [29] and, with their own classes rather than our control-based labels, for BRCA2 [30]. A certified level describes the calls as a class, so a borderline call can carry less evidence than its class. The guarantees assume residues exchangeable with the calibrated ones; they do not cover intermediate, splice-region or out-of-region variants, and per-class coverage after selection is guaranteed only at $1 - \max G$. Results cover five cancer-predisposition genes, four of them in DNA repair. REVEL was trained on labelled variants and the AlphaMissense classes were set on ClinVar, so both may overlap our evaluation sets [36, 40]; ESM-2, trained without variant labels, gave similar coverage and validity.

6 Conclusion

CertiVar turns a variant-effect predictor and a few hundred assayed residues of a gene into LoF, functional or no calls with finite-sample guarantees on per-class error rates (in expectation, at a fixed miscoverage level) and on call-averaged likelihood ratios with respect to the assay after threshold selection, without an estimate of how common damaging variants are. Code and data (release v1.2.3): `https://github.com/aaryavvatts-lab/certivar`.

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Appendix: supplementary material

Section, equation, figure and table numbers without the prefix S refer to the main paper. Contents: S1 proofs of Propositions 1–3, Theorems 1–2 and Corollary 1; S2 simulation (Fig. S1) and why marginal thresholds do not control the minority class; S3 calibration size, region and gene transfer (Fig. S2A–C), per-gene coverage (Table S1) and prior estimation (Table S2); S4 strata, assay-label sensitivity (Table S3) and calls judged against ClinVar labels (Table S4); S5 implementation; S6 experimental designs (Table S5) and all baselines by gene and predictor (Table S6); S7 further analyses: more comparison rules, PAC coverage without test-half noise, residue-weighted calibration, a check of class-conditional invariance and the number of VUS that receive certified calls (Tables S7–S10).

S1 Proofs

Assumptions, propositions, theorems and the corollary are stated in Section 3.1; Q, C, $\hat{t}_k$, L^+ and U^- are defined in Section 3.

Throughout, fix a class k and write F_k for the distribution function of the nonconformity r_k of a variant drawn from $Q(\cdot \mid Y = k)$. By Assumption 1, the kept calibration variants (one per residue of $\mathcal{A}$) and the variant to be called are independent draws from Q; conditionally on all labels, the n_k kept variants of class k and a class-k test variant are independent draws from $Q(\cdot \mid Y = k)$.

Proposition 1. Conditionally on the labels, the $n_k + 1$ nonconformities of the class-k kept variants and of the test variant are exchangeable, so with ties broken at random the rank of the test value is uniform on $\{1, \ldots, n_k + 1\}$ [12, 41]. The test value is at most the j_k-th smallest calibration value with probability $j_k/(n_k + 1)$, which lies in $[1 - \alpha_k,\ 1 - \alpha_k + 1/(n_k + 1))$ for $j_k = \lceil (n_k + 1)(1 - \alpha_k) \rceil$; ties only increase the probability, which gives the lower bound in general. If $j_k > n_k$ then $\hat{t}_k = \infty$. Since $k \in C$ exactly when $r_k \le \hat{t}_k$, the claim follows. □

Proposition 2. Conditionally on the labels, the coverage of the threshold is $F_k(\hat{t}_k) = F_k(r^{(k)}_{(j)})$. If F_k is continuous, $F_k(r^{(k)}_{(j)})$ is the j-th order statistic of n_k independent uniforms and has the $\mathrm{Beta}(j, n_k - j + 1)$ distribution, so $\Pr\{F_k(\hat{t}_k) < 1 - \alpha_k\} = F_{\mathrm{Beta}(j, n_k - j + 1)}(1 - \alpha_k) \le \delta$ by the choice of j_k [14]. If F_k has atoms, write $r = F_k^{-1}(U)$ with U uniform and F_k^{-1} the generalized inverse; then $r^{(k)}_{(j)} = F_k^{-1}(U_{(j)})$ and $F_k(F_k^{-1}(u)) \ge u$, so $F_k(r^{(k)}_{(j)}) \ge U_{(j)}$ and the bound still holds. If no $j \le n_k$ qualifies, $\hat{t}_k = \infty$ and coverage is 1. □

Proposition 3. Fix $\mathcal{A}$. Both ranks in (1) are non-increasing in α (the in-expectation rank because $\lceil (n_k + 1)(1-\alpha) \rceil$ is, the PAC rank because $F_{\mathrm{Beta}(j, n_k - j + 1)}(1-\alpha)$ decreases in α for each j, so the smallest qualifying j cannot increase), so $\hat{t}_k(\alpha)$ is non-increasing in α and the set $\{r_k \le \hat{t}_k(\alpha)\}$ shrinks as α grows. Hence, for any $\hat{\alpha} \in G$ that depends on $\mathcal{B}$ but not on the test variant, $\Pr(k \in C_{\hat{\alpha}} \mid Y = k, \mathcal{A}, \mathcal{B}) \ge \Pr(k \in C_{\max G} \mid Y = k, \mathcal{A})$, and taking expectations over $\mathcal{A}$ and Proposition 1 at $\alpha_k = \max G$ give the first claim. For the PAC rank at level $\delta/|G|$, Proposition 2 applied to each $\alpha \in G$ and a union bound give, with probability at least $1 - \delta$, coverage at least $1 - \alpha$ for every $\alpha \in G$ simultaneously, in particular for $\hat{\alpha}$. □

Theorem 1. The thresholds $\hat{t}(\alpha)$ depend only on $\mathcal{A}$ and are fixed given $\mathcal{A}$. Conditionally on $\mathcal{A}$ and on the labels of the kept variants of $\mathcal{B}$, the class-1 kept variants of $\mathcal{B}$ are n_1 independent draws from $Q(\cdot \mid Y = 1)$, so the number of them called LoF is $\mathrm{Binomial}(n_1, \mathrm{TP}(\alpha))$ with $\mathrm{TP}(\alpha) = \Pr(\text{call LoF} \mid Y = 1)$, and likewise for the other three counts. Each one-sided Clopper–Pearson bound [28] at level $\delta/(2|G|)$ fails with probability at most $\delta/(2|G|)$; the conventions for $x = 0$, $x = n$ and $n = 0$ in Section 3 give the trivial bounds 0 and 1, which never fail. For LR^+ there are $2|G|$ such bounds (ℓ_{TP} and u_{FP} for each α), so by the union bound all of them hold simultaneously with probability at least $1 - \delta$. On that event $\mathrm{TP}(\alpha) \ge \ell_{\mathrm{TP}}(\alpha)$ and $\mathrm{FP}(\alpha) \le u_{\mathrm{FP}}(\alpha)$ for every $\alpha \in G$, hence $\mathrm{LR}^+(\hat{t}(\alpha)) \ge L^+(\alpha)$ for every α (with $\mathrm{LR}^+ = \infty$ when $\mathrm{FP}(\alpha) = 0$), including any $\hat{\alpha}^+$ chosen from these data. Since this probability bound holds for every $\mathcal{A}$, it also holds unconditionally. The

argument for LR^- is identical with u_{FN} and ℓ_{TN}. The event does not involve the levels τ, so it covers every evidence level reported. □

Theorem 2. Fix $\mathcal{A}$, so the thresholds and the hypotheses $H_\alpha : \mathrm{LR}^+(\hat{t}(\alpha)) \le \tau$ are fixed. If H_α is true and both endpoints $\ell_{\mathrm{TP}}(\alpha)$ and $u_{\mathrm{FP}}(\alpha)$ at level $\delta/2$ hold, then $\ell_{\mathrm{TP}} \le \mathrm{TP} \le \tau\,\mathrm{FP} \le \tau\, u_{\mathrm{FP}}$, so H_α is not rejected; hence a true H_α is rejected with probability at most δ. Let α^* be the first α in the fixed order for which H_α is true (if none, every claim is true). The procedure claims the level at a false α only if it rejects every hypothesis up to that α, which includes H_{α^*}; this has probability at most δ. The selected α is one of the rejected ones, so the claimed level holds with probability at least $1-\delta$. The functional case is identical with $u_{\mathrm{FN}}/\ell_{\mathrm{TN}} < \tau'$. □

Corollary 1. Every quantity bounded in Propositions 1–3 and Theorems 1–2 is a probability conditional on the true class: $\Pr(k \in C \mid Y = k)$, the four call rates, and their ratios. Under Assumption 2 these are the same for target variants as for a test variant under Assumption 1, whatever π is, so Propositions 1–3 and Theorems 1–2 transfer. The two call sets are disjoint for any $\alpha^+, \alpha^- \in G$: a LoF call at α^+ requires $q > \hat{t}_0(\alpha^+)$ and $1-q \le \hat{t}_1(\alpha^+)$, a functional call at α^- requires $q \le \hat{t}_0(\alpha^-)$ and $1-q > \hat{t}_1(\alpha^-)$, and both thresholds are non-increasing in α, so the first conditions contradict each other if $\alpha^- \ge \alpha^+$ and the second if $\alpha^- < \alpha^+$. A functional call on a LoF variant requires $1 \notin C$, so its rate is at most $1 - \Pr(1 \in C \mid Y = 1) \le \alpha_1$. None of these quantities involves π, so no prior is needed to certify them; this concerns the likelihood ratio with respect to the assay outcome, not evidence about pathogenicity. □

S2 Simulation

Scores were drawn per residue as $s = 2y + u_c + e$, with a residue effect $u_c \sim N(0, 0.5^2)$ shared by the five variants of a residue, unit Gaussian noise, and the class shared within a residue. A Platt calibrator was fitted on 600 variants at LoF share 0.45, conformal thresholds were set on another 600 (one variant per residue), and 3,000 test variants were drawn at LoF shares from 0.05 to 0.60, with 300 replicates per share (Figure S1). Per-class thresholds kept LoF coverage at the target at every share; pooled thresholds did not, and reweighting them [26] to the EM estimate of the share [24] kept marginal coverage near the target while LoF coverage fell further as LoF variants became rarer. Adjusting posteriors to the EM estimate removed the calibration error caused by the shift in this simulation, where the shift is a pure change of prior, unlike in the real data (Section 4.2).

Why marginal thresholds do not control the minority class. A pooled split-conformal threshold $\hat{t}$ is a quantile of the mixture $\pi_c F_1 + (1-\pi_c)F_0$ of the two class-conditional nonconformity laws, weighted by the calibration share π_c. Its coverage of class 1 is $F_1(\hat{t})$, which is not controlled; when LoF variants are rare and harder to separate, $F_1(\hat{t}) < 1-\alpha$ while marginal coverage is still near $1-\alpha$. Reweighting to a known test prior [26] restores marginal coverage at that prior but does not control $F_1(\hat{t})$ either.

S3 Calibration size, region and gene transfer, and per-gene results

Figure S2 shows calibration size, prior estimates and calibration source; Table S1 gives per gene and predictor the residue-weighted coverage of each class and the abstention rate of the per-class rule and two baselines; Table S2 gives the estimated LoF shares and calibration errors behind Section 4.2. When the calibration and test halves of a gene share their LoF share, label-shift-weighted conformal prediction [26] with the true share nearly coincides with pooled conformal; with the EM estimate it inherits the estimation error, which collapsed its LoF coverage in some pairs (worst-class coverage as low as 0.039 at $\alpha = 0.1$).

S4 Strata, assay-label sensitivity and ClinVar check

For one within-gene split per gene and predictor ($\alpha = 0.1$, variant-weighted), coverage of LoF and functional variants was 0.966 and 0.929 on the third of variants farthest from the assay's class boundary, 0.929 and

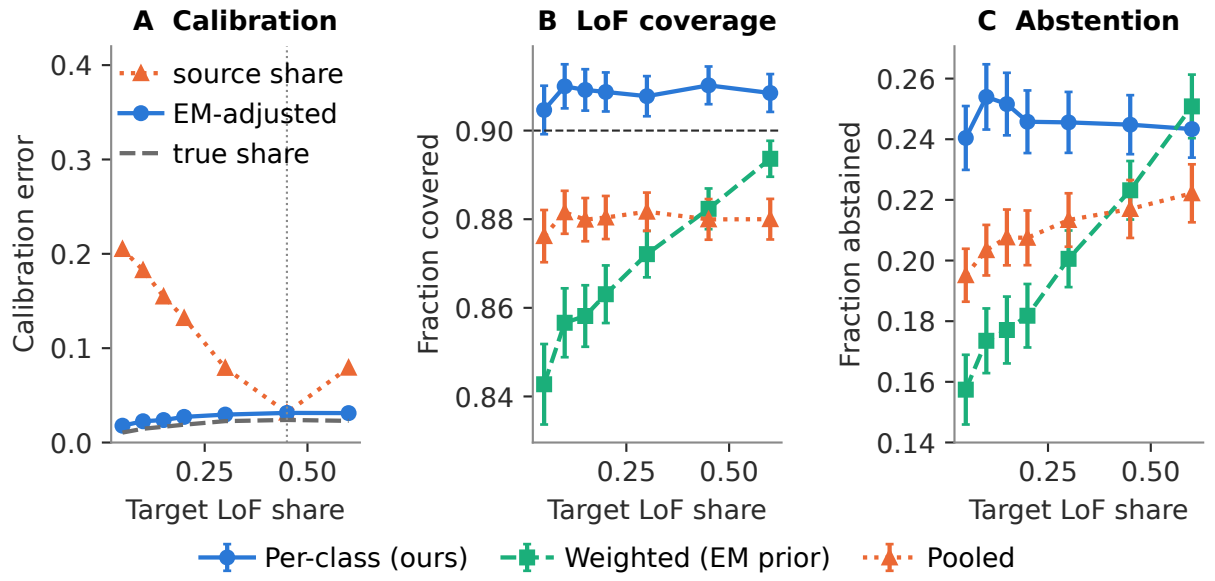


Figure S1: Simulation with residue-clustered scores, calibration at LoF share 0.45 and test share on the horizontal axis (300 replicates; error bars are 95% intervals of the mean). (A) Expected calibration error of Platt scaling at the source share, after EM prior adjustment, and after adjustment to the true share. (B) Coverage of LoF variants. (C) Abstention rate. Panel A has its own legend.

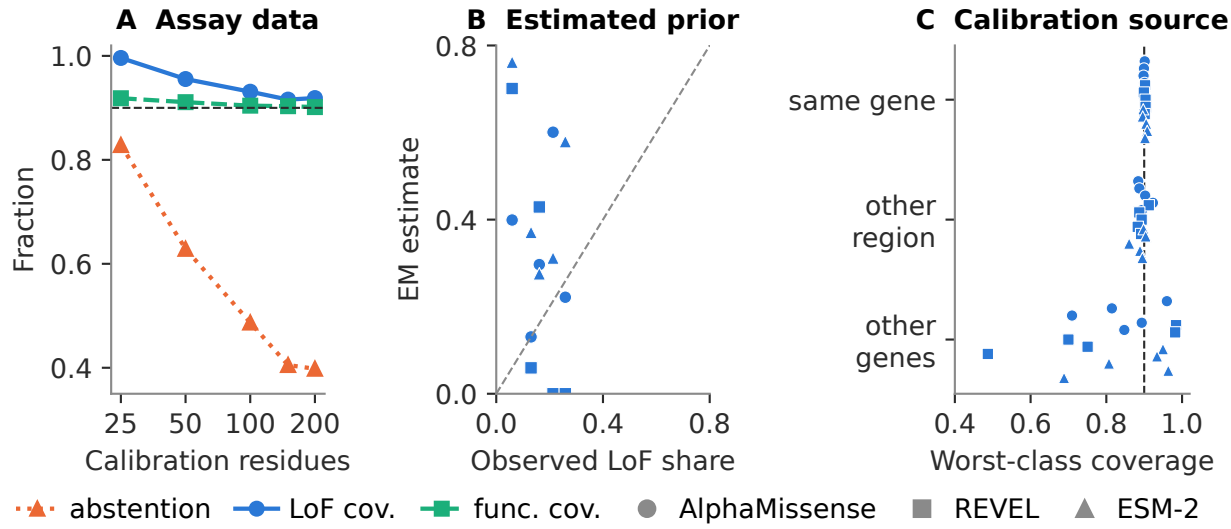


Figure S2: (A) Coverage and abstention against the number of calibration residues drawn from the gene (mean over genes, predictors and 50 draws). (B) EM estimate of the LoF share from unlabelled scores against the observed share; the dashed line is equality. (C) Worst-class coverage when calibrating on random residues of the same gene, on the other three quarters of the protein, or on the other four genes.

0.912 on the middle third, 0.865 and 0.873 on the nearest third, and 0.881 and 0.896 on variants classified VUS in ClinVar, with abstention rates of 0.334, 0.385, 0.413 and 0.392.

Table S3 gives, for each label rule, the number of labelled variants, their LoF share, the coverage and abstention of the per-class rule, and the oracle check of the certified selection (Section 4.5). This analysis is an independent run with its own random splits, so its main-label rows differ from Tables S1 and S6 by Monte Carlo error (for RAD51C with ESM-2, LoF coverage 0.895 and abstention 0.28 here against 0.915 and 0.34 in Table S1).

Table S1: Within-gene random split, $\alpha = 0.1$, mean of 50 splits. Each cell is residue-weighted LoF coverage / functional coverage / abstention rate.

Predictor	Gene	Per-class (ours)	Pooled conformal	Fixed 0.1/0.9
AlphaMissense	BRCA1	0.916/0.901/0.25	0.952/0.889/0.32	0.971/0.927/0.44
AlphaMissense	BRCA2	0.899/0.903/0.36	0.808/0.921/0.17	0.890/0.965/0.37
AlphaMissense	BAP1	0.909/0.898/0.39	0.977/0.880/0.69	0.942/0.687/0.26
AlphaMissense	RAD51C	0.901/0.905/0.17	0.824/0.929/0.10	0.961/0.973/0.37
AlphaMissense	MSH2	0.915/0.901/0.29	0.972/0.897/0.70	0.956/0.799/0.39
REVEL	BRCA1	0.914/0.902/0.34	0.962/0.883/0.48	0.997/0.999/0.98
REVEL	BRCA2	0.912/0.899/0.42	0.841/0.911/0.24	0.947/0.998/0.68
REVEL	BAP1	0.903/0.904/0.42	0.545/0.997/0.15	0.663/1.000/0.24
REVEL	RAD51C	0.909/0.904/0.27	0.625/0.999/0.20	0.680/1.000/0.23
REVEL	MSH2	0.911/0.901/0.37	0.991/0.896/0.79	0.991/0.900/0.80
ESM-2	BRCA1	0.911/0.897/0.37	0.912/0.893/0.34	0.970/0.975/0.64
ESM-2	BRCA2	0.896/0.903/0.36	0.944/0.894/0.51	0.964/0.958/0.71
ESM-2	BAP1	0.910/0.905/0.47	0.878/0.912/0.36	0.903/0.943/0.46
ESM-2	RAD51C	0.915/0.908/0.34	0.973/0.886/0.50	0.987/0.963/0.71
ESM-2	MSH2	0.920/0.902/0.46	1.000/0.895/0.80	0.993/0.823/0.68

Table S2: Prior estimation and calibration on assayed variants outside the ClinVar labelled set. LoF share observed and estimated from unlabelled scores by expectation–maximization (EM) [24] and black-box shift estimation (BBSE) [25]; expected calibration error (10 equal-mass bins) of the ClinVar-Platt posterior, after EM prior adjustment, after adjustment to the true prior, and of Platt scaling fitted on the other four genes' assay labels. AM: AlphaMissense.

		LoF share			Expected calibration error			
Pred.	Gene	obs.	EM	BBSE	ClinVar Platt	+ EM	+ true prior	other genes
AM	BRCA1	0.161	0.297	0.316	0.203	0.135	0.064	0.049
AM	BRCA2	0.130	0.131	0.178	0.132	0.032	0.032	0.041
AM	BAP1	0.213	0.601	0.588	0.341	0.388	0.255	0.119
AM	RAD51C	0.259	0.222	0.252	0.067	0.043	0.037	0.162
AM	MSH2	0.060	0.399	0.395	0.361	0.340	0.120	0.139
REVEL	BRCA1	0.161	0.429	0.336	0.285	0.268	0.058	0.140
REVEL	BRCA2	0.130	0.059	0.149	0.176	0.071	0.025	0.111
REVEL	BAP1	0.213	0.000	0.000	0.092	0.213	0.149	0.138
REVEL	RAD51C	0.259	0.000	0.000	0.143	0.259	0.185	0.187
REVEL	MSH2	0.060	0.701	0.577	0.485	0.642	0.106	0.329
ESM-2	BRCA1	0.161	0.276	0.304	0.212	0.114	0.050	0.044
ESM-2	BRCA2	0.130	0.371	0.371	0.288	0.241	0.075	0.035
ESM-2	BAP1	0.213	0.312	0.327	0.171	0.121	0.086	0.092
ESM-2	RAD51C	0.259	0.579	0.538	0.246	0.320	0.122	0.102
ESM-2	MSH2	0.060	0.762	0.670	0.531	0.702	0.165	0.234

Table S4 gives the exploratory check of the calls against ClinVar labels (Section 4.4), per gene and pooled. Only BRCA1, BRCA2 and MSH2 have enough labelled residues for informative bounds; across them the LoF-call lower bounds differ by gene, so the pooled values should not be read as a common likelihood ratio. Some ClinVar classifications already use these assays or predictors as evidence, so this check is not independent clinical validation.

Table S3: Sensitivity to the assay-label rule. Each gene is relabelled with the authors' classes or with the control rule at the stated posterior cut-off; other genes are unchanged. Per-class: lowest mean per-class coverage over the three predictors (50 random halvings, $\alpha = 0.1$); abstention: range over predictors; bounds held: oracle selections whose certified bound the true likelihood ratio respected (100 draws per predictor); LoF-supp.: share of draws certifying LoF-supporting calls, range over predictors.

Gene	Labels	n	LoF	Per-class	Abstention	Bounds held	LoF-supp.
BRCA2	controls, 0.80	3,971	0.136	0.894	0.49–0.53	294/295	0.63–0.96
BRCA2	controls, 0.95 (main)	3,435	0.130	0.896	0.36–0.42	423/423	0.90–0.99
BRCA2	controls, 0.99	2,341	0.151	0.897	0.30–0.37	591/593	0.94–1.00
BAP1	authors (main)	4,294	0.213	0.901	0.39–0.47	460/460	0.90–1.00
BAP1	controls, 0.95	4,221	0.100	0.899	0.13–0.19	777/777	1.00
RAD51C	authors (main)	2,225	0.259	0.895	0.18–0.28	558/561	0.84–0.99
RAD51C	controls, 0.95	832	0.538	0.902	0.07–0.19	868/868	0.98–1.00

Table S4: Calls judged against ClinVar labels (exploratory). Thresholds ($\alpha = 0.1$) are set on each gene's assay labels; one ClinVar variant with $\geq$2 stars is kept per residue. n: kept P/LP and B/LB variants; calls: LoF (or functional) calls among P/LP / among B/LB variants; L^+, U^-: likelihood-ratio bounds from one-sided Clopper–Pearson endpoints at level 0.025 each (simultaneous level 0.95 per ratio). Pooled rows sum the kept counts over genes, each gene with its own thresholds. BAP1 and RAD51C have too few labelled residues to be informative.

Pred.	Gene	n P/LP	n B/LB	LoF calls	L^+	func. calls	U^-
AM	BRCA1	75	170	56/1	19.6	8/153	0.24
AM	BRCA2	60	210	49/8	9.44	6/164	0.29
AM	BAP1	6	8	5/0	0.97	1/8	1.02
AM	RAD51C	8	3	5/1	0.27	2/2	6.90
AM	MSH2	72	55	50/6	2.58	9/46	0.31
AM	pooled	221	446	165/16	11.9	26/373	0.21
REVEL	BRCA1	75	170	48/10	4.94	8/134	0.28
REVEL	BRCA2	60	210	44/9	7.56	11/182	0.37
REVEL	BAP1	6	8	4/0	0.60	2/8	1.23
REVEL	RAD51C	8	3	6/0	0.49	2/3	2.23
REVEL	MSH2	72	55	43/3	3.14	5/38	0.28
REVEL	pooled	221	446	145/22	7.99	28/365	0.23
ESM-2	BRCA1	75	170	47/1	15.7	10/159	0.26
ESM-2	BRCA2	60	210	33/4	8.66	9/184	0.32
ESM-2	BAP1	6	8	5/0	0.97	1/5	2.62
ESM-2	RAD51C	8	3	5/0	0.35	1/1	62.7
ESM-2	MSH2	72	55	44/3	3.23	3/33	0.25
ESM-2	pooled	221	446	134/8	15.4	24/382	0.19

S5 Implementation

Inputs: ClinVar `variant_summary.txt.gz` of 26 October 2026 (SHA-256 in `data/MANIFEST.txt`); MaveDB score sets `urn:mavedb:00000097-0-2` (BRCA1), `00001279-a-1` (BRCA2), `00000662-0-1` (BAP1), `00000673-0-1` (RAD51C) and `00000050-a-1` (MSH2, positive scores = loss of function); Ensembl VEP REST on GRCh38 (release 116) with the AlphaMissense and REVEL annotations; ESM-2 `esm2_t33_650M_UR50D`. Distance from the class boundary is the absolute difference between a variant's assay score and the midpoint of the two class medians in its gene, divided by the distance between the medians. Expected calibration error uses ten equal-mass bins. AUROC intervals are percentiles of 1,000 bootstrap resamples of residues. The region split divides each gene's assayed residues into four contiguous blocks with equal numbers of distinct positions. ClinVar records whose aggregate germline classification is empty (somatic-only submissions) are treated as unclassified. All random draws use NumPy generators with

fixed seeds (2027 plus a fixed offset per experiment). The experiments run on a laptop CPU with Python 3.13, NumPy 2.5, SciPy 1.18, scikit-learn 1.9 and pandas 3.1; ESM-2 scoring used fair-esm 2.0 and PyTorch 2.14 on an Apple-silicon GPU. Exact versions and data checksums are in the repository. The Platt map is checked to be increasing (positive slope) every time it is fitted, and every fit in the experiments passed. Certified bounds use the conventions of Section 3: $U^- = \infty$ when $\ell_{\mathrm{TN}} = 0$, an empirical ratio $x/0$ is ∞ for $x > 0$, and $0/0$ is undefined (never a claim); boundary cases are covered by unit tests (`make test`). The VUS-shift LoF shares are residue-weighted for both the calibration sample (one variant per residue) and the VUS (each residue weighted equally, then its variants).

S6 Experimental designs and all baselines

Table S5 lists the designs of the main text; every split is by residue, and calibration keeps one variant per residue. Table S6 gives, for every gene and predictor, the per-class rule next to pooled conformal prediction and the published cut-offs.

Table S5: Experimental designs.

Experiment	Residue split	Tests	Reps
Coverage (4.2)	halves: calibrate on one, test on the other	Prop. 1	50
Calibration share (4.2)	halves; calibration half subsampled to LoF share 0.05–0.5	Cor. 1	100
Reliability (4.3)	halves	Props. 1–2	200
VUS (4.3)	halves; non-VUS of one half (40% calibrate, 60% certify); VUS of the other	Cor. 1	50
Oracle (4.4)	30% and 50% of residues drawn with replacement; truth on all residues	Thms. 1–2, Prop. 3	100
Held-out fifth (4.4)	disjoint 30% / 50% / 20%	Thm. 1	100
Regions (4.5)	three contiguous quarters / the fourth	beyond Asm. 1	25/quarter

S7 Further analyses

These analyses (`code/18_robustness.py` and `code/19_baselines.py`) answer questions the main text leaves open; they use their own random draws, so their numbers for the main rules differ slightly from Section 4.

More comparison rules. Table S7 repeats the oracle design of Section 4.4 with Katz intervals for a ratio of binomial proportions [22] at level $\delta/|G|$ per ratio (0.5 added to every cell of a ratio with a zero count), the percentile bootstrap at the Bonferroni level, the uncorrected percentile bootstrap with 0.5 added to every cell, and fixed-sequence testing with each of the three levels of a call type at $\delta/3$, which holds for all levels at once (Theorem 2 and a union bound over levels). Every claimed bound of a draw held in 99.7% of 1,500 draws for the certified rule, 97.9% for uncorrected Clopper–Pearson bounds and 98.0% for Katz intervals, against 57.6%, 65.3% and 75.4% for the three bootstrap variants. Katz intervals have no finite-sample guarantee but held here and certified LoF-moderate in 52.5% of draws against 13.3%, so the exact certificate costs yield in exchange for a guarantee at every sample size. With the same simultaneous guarantee as the Bonferroni rule, fixed-sequence testing traded supporting for moderate evidence: LoF-moderate in 14.4% of draws against 13.3% and LoF-supporting in 74.9% against 86.7% on G_4, and LoF-moderate in 12.7% against 6.4% and LoF-supporting in 50.3% against 74.8% on the 20-point grid.

PAC coverage without test-half noise. For 200 calibration samples per pair drawn with replacement from a gene's residues, we computed per-class coverage of the thresholds exactly on all of the gene's residues, the quantity Proposition 2 bounds. PAC thresholds ($\delta_{\mathrm{P}} = 0.1$) fell below 0.90 in 4.8% (LoF) and 7.1%

Table S6: Calls by gene and predictor ($\alpha = 0.1$; calibration and test on disjoint halves of the gene's residues, mean of 50 splits). Per-class rule: residue-weighted coverage of LoF and functional (Fun) variants and abstention rate (Abs); Monte Carlo standard errors of the coverages are 0.002–0.010. Baselines: worst-class coverage (bold if below 0.85) and abstention. Published cut-offs: AlphaMissense classes (AM) and REVEL PP3/BP4 thresholds; none exist for ESM-2.

		Per-class (ours)			Pooled conformal		Published cut-off	
Gene	Predictor	LoF	Fun	Abs	Worst	Abs	Worst	Abs
BRCA1	AM	0.916	0.901	0.25	0.889	0.32	**0.773**	0.13
	REVEL	0.914	0.902	0.34	0.883	0.48	**0.663**	0.59
	ESM-2	0.911	0.897	0.37	0.893	0.34	–	–
BRCA2	AM	0.899	0.903	0.36	**0.808**	0.17	**0.810**	0.11
	REVEL	0.912	0.899	0.42	**0.841**	0.24	**0.803**	0.57
	ESM-2	0.896	0.903	0.36	0.894	0.51	–	–
BAP1	AM	0.909	0.898	0.39	0.880	0.69	**0.540**	0.07
	REVEL	0.903	0.904	0.42	**0.545**	0.15	**0.799**	0.29
	ESM-2	0.910	0.905	0.47	0.878	0.36	–	–
RAD51C	AM	0.901	0.905	0.17	**0.824**	0.10	0.860	0.11
	REVEL	0.909	0.904	0.27	**0.625**	0.20	**0.792**	0.25
	ESM-2	0.915	0.908	0.34	0.886	0.50	–	–
MSH2	AM	0.915	0.901	0.29	0.897	0.70	**0.642**	0.12
	REVEL	0.911	0.901	0.37	0.896	0.79	**0.445**	0.41
	ESM-2	0.920	0.902	0.46	0.895	0.80	–	–

Table S7: All selection rules on the same 1,500 oracle draws (G_4, $\delta = 0.05$; a run separate from Section 4.4). Guarantee after selection: for all evidence levels of a call type at once, per level, or none. Bounds held, levels held: draws in which every claimed bound (fixed sequence certifies levels only; a point estimate is no bound) or level held (%). Last four columns: draws certifying each level (%).

	Guarantee	Bounds	Levels	LoF calls		Functional calls	
Selection rule	after selection	held	held	supp.	mod.	supp.	mod.
Certified, Bonferroni (ours)	all levels	99.7	99.9	86.7	13.3	44.9	1.5
Fixed sequence, levels at $\delta/3$	all levels	–	99.9	74.9	14.4	43.8	1.5
Fixed sequence, levels at δ	per level	–	99.7	87.2	30.5	61.9	3.9
Clopper–Pearson, uncorrected	none	97.9	99.5	95.9	35.1	68.0	6.9
Katz interval, Bonferroni	none	98.0	99.7	99.3	52.5	52.4	0.2
Percentile bootstrap	none	57.6	72.5	99.9	78.1	90.0	45.1
Percentile bootstrap, Bonferroni	none	65.3	75.1	98.1	52.3	74.3	36.8
Percentile bootstrap, +0.5 counts	none	75.4	88.2	99.9	76.6	89.3	37.3
Point estimate	none	–	42.1	100.0	98.2	99.8	75.7

(functional) of draws and in at most 10.0% in any pair, within δ_{P}; in-expectation thresholds did so in 34.3% and 42.9%. The 10.8% of Section 4.3 therefore reflects the sampling noise of a finite test half.

All variants with residue weights. In a separate run of 50 halvings, calibrating on all variants of the calibration half left worst-class coverage below 0.90 in 12 of 15 pairs (lowest 0.885), against 5 (lowest 0.897) for one variant per residue. Weighting each variant by one over the number of same-class calibration variants at its residue, so that every residue counts once as under Q, left 1 pair below 0.90 (lowest 0.898) but abstained on 44.6% of variants against 35.5% for one variant per residue.

Checking class-conditional invariance. Under Assumption 2 the target score law is a mixture $\pi F_1 + (1-\pi)F_0$ of the two class laws. We test this necessary condition on one target variant per residue with

$D = \min_\pi \sup_x |\hat{F}_T(x) - \pi\hat{F}_1(x) - (1-\pi)\hat{F}_0(x)|$ (π on 201 values in $[0,1]$), where $\hat{F}_1$ and $\hat{F}_0$ are the empirical class distribution functions of the calibration data; the p-value comes from 300 bootstrap resamples that redraw both classes and a target sample of the same size from the fitted mixture. A rejection shows that the targets are not a reweighting of the calibration classes; acceptance does not establish Assumption 2. The check rejected 4.7% of random halves (3.0–7.7% by gene), where the assumption holds by design, 43.9% of held-out contiguous quarters, 77.3% of transfers from another gene and every tested set of unassayed ClinVar VUS at residues the BRCA1 and BRCA2 assays did not reach, but only 8.5% of assayed-VUS targets (Table S8); except for random halves, where the two were equal, rejected targets had lower coverage than accepted ones.

Table S8: Mixture-consistency check of Assumption 2 ($p < 0.05$; one target variant per residue, 300 bootstrap resamples). Coverage: mean over target sets of the smaller per-class coverage in each set of in-expectation thresholds ($\alpha = 0.1$) set on the same calibration data, where labels exist.

		Target	Rejected	Per-set worst-class coverage	
Target set	Assumption 2	sets	(%)	accepted	rejected
Random half of the residues	holds by design	1,500	4.7	0.880	0.880
Assayed VUS at other residues	approximately	750	8.5	0.845	0.816
Held-out contiguous quarter	not assumed	1,500	43.9	0.837	0.788
Another gene's calibration data	not assumed	300	77.3	0.745	0.644
Unassayed VUS, other residues	unknown (no labels)	120	100.0	–	–

How many VUS receive certified calls. Of the 8,909 ClinVar missense VUS in the five genes, 147 are unassayed but lie at assayed residues, 105 of them in MSH2 (Table S9). For MSH2 the check rejected 40% of samples with AlphaMissense and 35% with REVEL but none with ESM-2, whose thresholds gave certified LoF-supporting calls to 19.1 and functional-supporting calls to 5.0 of the 105 on average (Table S10). The 4,010 unassayed BRCA1 and BRCA2 VUS at other residues are outside what the assays can certify: the check rejected every sample of them.

Table S9: ClinVar missense VUS by assay status. Assayed: VUS with an assay score, including intermediate and splice-region variants (Table 1 counts only those with a binary label). Check: share of 20 mixture checks rejected ($p < 0.05$) for the unassayed VUS at other residues, run where they span at least 30 residues.

			Unassayed		Check rejected,
Gene	All	Assayed	at assayed residues	elsewhere	elsewhere (%)
BRCA1	1,909	375	29	1,505	100.0
BRCA2	3,508	1,003	0	2,505	100.0
BAP1	1,189	1,173	3	13	–
RAD51C	596	570	10	16	–
MSH2	1,707	1,597	105	5	–
Total	8,909	4,718	147	4,044	

Table S10: Certified calls on the ClinVar VUS that are unassayed but lie at assayed residues (Algorithm 1, G_4, $\delta = 0.05$; mean number of calls over 20 random 40/60 calibration/certification splits of all assayed residues), and the share of mixture checks of those VUS rejected ($p < 0.05$), run where they span at least 30 residues; where the check was run, calls are averaged over the splits it accepted.

			LoF calls		Functional calls		Check
Gene	VUS	Predictor	supp.	mod.	supp.	mod.	rejected (%)
BRCA1	29	AlphaMissense	7.9	0.2	11.7	0.0	–
		REVEL	7.2	0.0	3.8	0.0	–
		ESM-2	8.0	1.0	5.0	0.0	–
BAP1	3	AlphaMissense	2.9	0.6	0.0	0.0	–
		REVEL	2.7	0.4	0.0	0.0	–
		ESM-2	2.9	0.0	0.0	0.0	–
RAD51C	10	AlphaMissense	2.8	0.8	5.7	1.2	–
		REVEL	3.9	0.5	3.4	0.0	–
		ESM-2	3.4	0.1	4.2	0.0	–
MSH2	105	AlphaMissense	31.8	5.6	60.8	4.8	40.0
		REVEL	20.0	0.0	22.9	0.0	35.0
		ESM-2	19.1	0.8	5.0	0.0	0.0